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PHYSICAL CHEMISTRY

Calcium Carbonate and Titanium Dioxide Particles as a Basis for Container Fabrication for Brain Delivery of Compounds

T. V. Bukreeva[a], I. V. Marchenko[b], T. N. Borodina[a], I. V. Degtev[a], S. L. Sitnikov[b], Yu. V. Moiseeva[c], N. V. Gulyaeva[c], and Corresponding Member of the RAS M. V. Kovalchuk[a]

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The targeted drug delivery to brain is an important problem both for fundamental science and for medical applications. The lack of solution to this problem considerably restricts the development of modern approaches to the diagnosis and treatment of cerebral pathology. Currently, of considerable interest of researchers is the intranasal route of administration of a number of drugs (first of all, peptides). It is believed that intranasally administered drugs can directly penetrate the central nervous system (CNS) through the olfactory system bypassing the brain–blood barrier [1]. The advantages of this administration route are obvious: the introduction is non-invasive, and the bioavailable drug is rapidly absorbed and starts to function as soon as after several minutes [2], which is an important point for emergency aid.

Of course, not any substances can be efficiently administered by this route but extending the range of drugs introduced in this way is undoubtedly topical and demanded. Good nasal bioavailability is known for lipophilic compounds with low molecular weight [3]. Peptides and proteins with molecular weight higher than 1000 Da are able to penetrate the nasal membrane only in small amounts [4]. Moreover, the clearly pronounced defense mechanism, namely, cleaning by the ciliated epithelium, restricts the time that liquids or powders can spend in the nasal cavity. The half-period of removal is 15–30 min [5]. To decrease the cleaning rate and increase the bioavailability of drugs, special absorption enhancers are used. Their action mainly consists in modification of the epithelial cell membrane permeability by modifying the phospholipid bilayer, removal of the proteins from the membrane or even elimination of the external layer of the mucous membrane [2].

[a] *Shubnikov Institute of Crystallography, Russian Academy of Sciences, Leninskii pr. 59, Moscow, 119991 Russia*
[b] *Moscow State University, Moscow, 119991 Russia*
[c] *Institute of Higher Nervous Activity and Neurophysiology, Russian Academy of Sciences, ul. Butlerova 5a, Moscow, Russia*

This study proposes a new system of brain delivery of functional compounds upon their intranasal administration, namely, the use of containers based on porous inorganic compounds.

As the basis for the containers for the delivery of compounds to the CNS, we used porous inorganic calcium carbonate and titanium dioxide microparticles. These particles have high adsorbability to a broad range of agents. The porous structure of these compounds ensures a high degree of loading of the container by the functional agents. Recently it was found that nano-sized titanium dioxide particles cause a strong additional neural inflammatory response in brain under conditions of preliminarily induced neural inflammation [6]. Therefore, it is of prime importance that the micron containers we propose do not allow the particles to penetrate into brain but ensure the penetration of only functional molecules and, hence, they are potentially safe.

The spherical $CaCO_3$ microparticles were obtained by mixing solutions of $CaCl_2$ and Na_2CO_3 by a reported procedure [7]. The particles had rather narrow size distribution (4 to 6 μm) and mesoporous structure (Fig. 1). The mesostructured titanium dioxide particles were prepared by the reaction of a titanium carbide powder with nitric acid [8]. The particles obtained in this way had an average size of 1.5 μm and consisted of elongated nanocrystallites (Fig. 2).

As the model drug for the delivery to the rat CNS, we used the central anesthetic loperamide, which is known to be unable to penetrate the brain–blood barrier. Loperamide was adsorbed on calcium carbonate and titanium dioxide particles from an ethanol solution. The amount of loperamide incorporated in the containers was determined by spectrophotometry. To increase the residence time of the containers in the nasal cavity, the particle surface was modified by hyaluronic acid having mucoadhesive properties. For this purpose, the loperamide-loaded particles were placed in a solution of hyaluronic acid in brine and agitated by a shaker. The particles were stored dry.

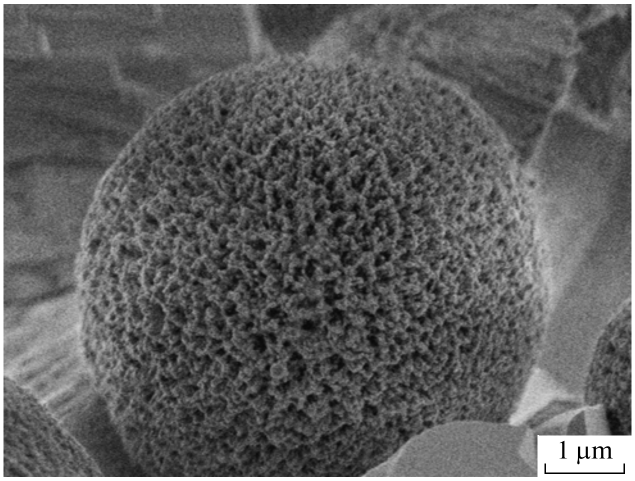

Fig. 1. Scanning electron microscopic image of calcium carbonate microparticles.

Previously, the addressed delivery of loperamide to the mouse CNS was successfully accomplished by intravenous injection of poly(butyl cyanoacrylate) nanoparticles [9]. The poly(alkyl cyanoacrylate) nanoparticles are currently among the best developed and effective transporting systems for the brain delivery of functional compounds [10]. In this study, loperamide-loaded poly(butyl cyanoacrylate) particles were administered intranasally, and the data on the efficiency of loperamide administration by means of calcium carbonate and titanium dioxide were compared with the data for polymeric particles.

A commonly accepted test for estimation of the drug delivery to brain is the formalin test using loperamide as the drug [11–13]. The conclusions about the loperamide delivery to the rodent CNS are drawn on the basis of changes in their pain sensitivity. A suspension of loperamide-loaded particles was administered intranasally in Wistar rats weighing 200–250 g. Then, after subcutaneous introduction of formaldehyde into the rat paw, the animal was placed into a transparent box for observation. After a specified period of time, the rat behavior was recorded and the pain sensitivity was scored in points. The time instants were averaged over the group and the results were processed using the STATISTICA-7 program. A loperamide solution in a 5% solution of glucose without particles and suspensions of particles containing no loperamide were used as controls. The pain sensitivity points of the control groups of animals did not differ significantly from each other; therefore, they were combined and the averaged control value was used in further calculations.

All of the loperamide-loaded particles we used did reduce the pain sensitivity in rats (Fig. 3). This means that the containers based on porous inorganic particles, as well as the poly(butyl cyanoacrylate) particles, deliver loperamide to the rat CNS. It can be seen in the figure that the patterns of decrease in the pain sensitivity are different for different types of particles. Titanium dioxide particles, like the poly(butyl cyanoacrylate) particles, are most efficient for short operation times, although the polymeric containers start to act sooner. Calcium carbonate particles provided a considerable decrease in the pain sensitivity at longer operation times (42–60 min).

The modification of particles by hyaluronic acid increases almost twice the efficiency of container operation in the beginning of the test and after long use (Fig. 4). Since at longer times, $CaCO_3$ particles loaded with loperamide are more efficient than poly(butyl cyanoacrylate) particles, one can conclude that calcium carbonate-based containers coated by hyaluronic acid is the system of choice among the systems studied for the delivery of loperamide to the rat CNS.

Thus, the results of the in vivo test demonstrated the high efficiency of the containers based on porous inorganic particles for the brain delivery of functional compounds upon their intranasal administration.

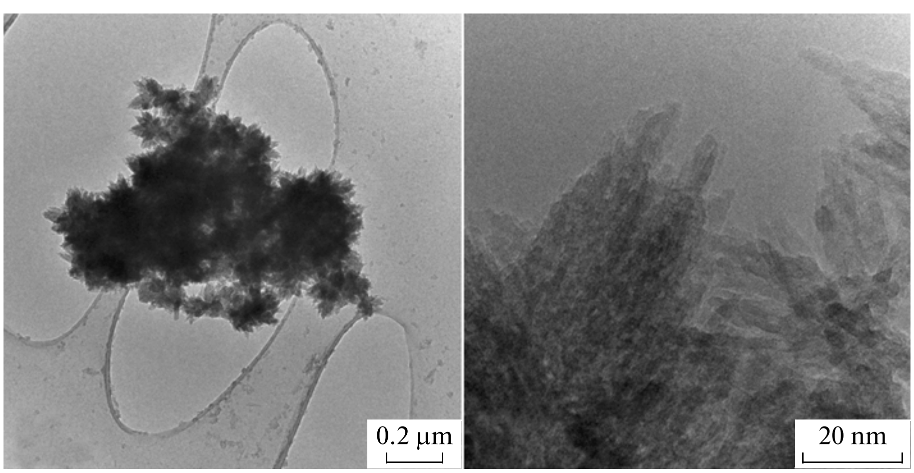

Fig. 2. Transmission electron microscopic images of titanium dioxide particles (with different magnifications).

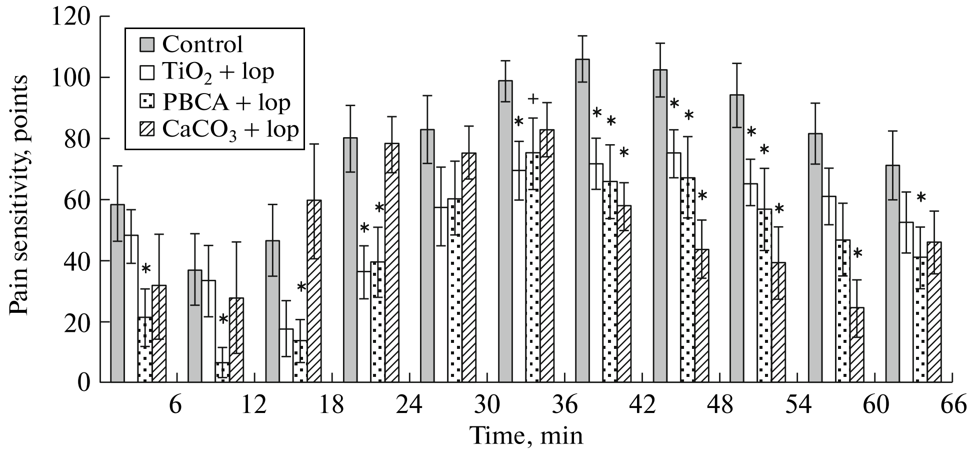

Fig. 3. Results of in vivo test of loperamide-loaded titanium dioxide, poly(butyl cyanoacrylate) (PBCA), and calcium carbonate particles. (*) Significant differences from the control group at (*) $p < 0.05$ and (+) $p < 0.1$; p is the statistical significance of the result according to the Kruskal–Wallis criterion.

The next important task is to estimate the toxicity of the containers. This will be done before the introduction of the containers into clinical practice. This task was beyond the scope of our work, nevertheless, there are some considerations concerning the exceptionally low toxicity of these containers. They are composed of nearly nontoxic substances, calcium carbonate and hyaluronic acid (these substances are present in considerable amounts in human organs and tissues) and titanium dioxide (used, in particular, as a food additive). Moreover, only a small amount of such particles can be introduced intranasally. The high efficiency and very low toxicity of the containers in combination with the preparation simplicity and the advantages of the intranasal administration route ensure the good prospects of the proposed system for medical applications.

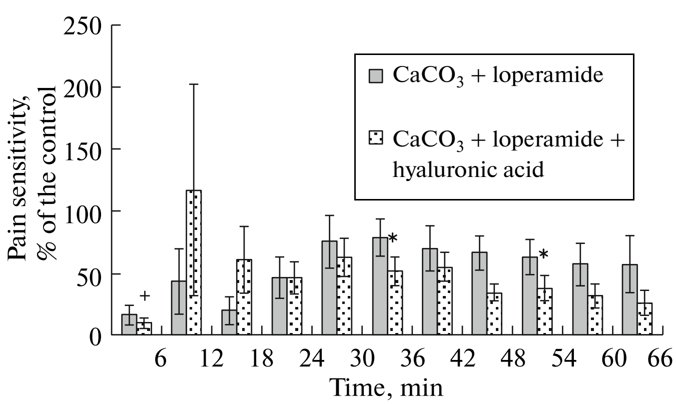

Fig. 4. Results of in vivo test of loperamide-loaded nonmodified calcium carbonate particles and the same particles coated by hyaluronic acid.

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